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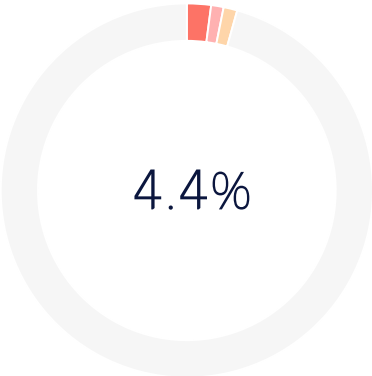

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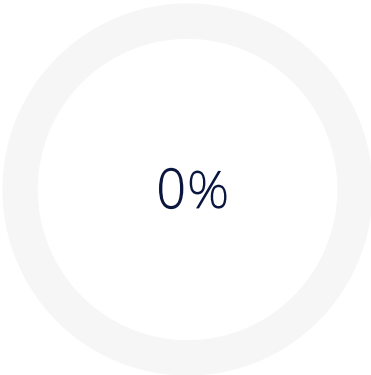

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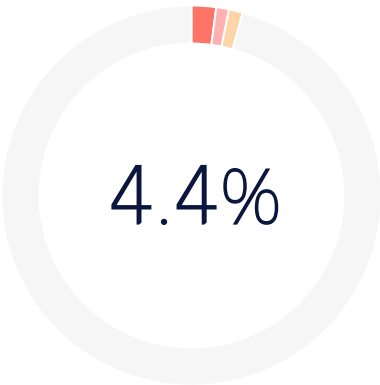

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Impact of Content Recommendation Systems on User Polarization

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***Abstract*—In an increasingly digitalized world, social media has established itself as the main channel of communication and information. The use of content recommendations on social media platforms has allowed the experience of each individual to be tailored according to their tastes and prior engagement with the platform. Although the use of these recommendations improves user participation on social media platforms, they can also create problems, such as the formation of echo chambers and user polarization. To determine how recommendation systems affect users' opinions and behaviour, we analysed user-generated data collected from Reddit and Twitter. We first removed any extraneous information from the data and processed it using basic natural language processing techniques. Next, we performed sentiment analysis to determine users' opinions. In addition to examining users' opinions, we categorised the two types of recommendation systems used on the study platforms into engagement-based and diversity-based systems. Our analysis indicated that users who utilised the engagement-based recommendation system had an increased likelihood of creating an echo chamber for themselves through the repeated viewing of similar content, whereas users who utilised diversity-based systems had decreased odds of being polarized due to being exposed to different points of view. These results underscore the need for careful design of recommendation systems to achieve a balance between maximising engagement and providing a diverse source of information to users.**

***Index Terms*—Content Recommendation Algorithms, User Polarization, Echo Chambers, Machine Learning, Network Analysis**

I. Introduction

With the popularity of mobile internet and the development of social media platforms, platforms like Twitter, Reddit, Facebook, etc. have emerged as the leading channels for communication and obtaining information. These platforms quickly attract a large audience with their interactive features, diverse multimedia content, and personalised user experience. However, these platforms position themselves as potent instruments in changing what people think . These platforms use content recommendation algorithms that allow the user's experience to be tailored according to their likes and prior engagements on the platform. Therefore, studying how recommendation systems influence user behaviour is highly significant in terms of practical contexts.

The core of the recommendation system is to analyse the user's viewing habits, interactive behaviour such as likes, comments, shares, etc. and personal preferences through data search and machine learning so that it can recommend relevant content to users. Studies have shown that these recommendation algorithms can remarkably increase the user's viewing time and user retention. [1] For example, research on YouTube shows that over 70% of viewing time comes from recommended videos. This demonstrates the strong influence of recommendation algorithms on watch time and user retention. Additionally, some studies found that the recommendation algorithm can keep improving its accuracy by constantly learning from users' behavioural data. This can lead to higher user satisfaction and more frequent interactions.

Different types of content can affect how recommendation algorithms influence users' opinions and interactions. Studies show that content that aligns with users' beliefs and is emotionally charged is more likely to grab attention and boost engagement. This type of content often reinforces users' existing beliefs and raises the chances of interacting with similar material. Content that presents different or opposing viewpoints can provide valuable information, but it often receives less attention and fewer recommendations from algorithms focused on engagement. Research indicates that algorithms tend to prioritise content that gets a lot of interaction. This can shape how users engage and often leads to people clustering around similar ideas, which contributes to formation of echo

chambers. Therefore, the type of content recommended is key to whether user behaviour drives more polarisation or encourages exposure to various perspectives.

Most of these recommendation algorithms tend to be based on engagement-based strategies, which emphasise metrics such as likes, comments, shares, and watch time. YouTube, TikTok, and Instagram are some of the platforms that use engagement-based algorithms to increase user retention. This approach may increase user engagement, but it may also strengthen existing beliefs and opinions, which may eventually lead to echo chambers and increased polarisation of ideologies.

On the other hand, diversity-aware recommendation strategies are aimed at providing users with a wider range of viewpoints by ensuring a variety of viewpoints or even contradictory viewpoints are included in the recommended feeds. Some platforms and information systems have started to explore diversity-aware strategies to provide more balanced information and minimise algorithmic bias. Additionally, some modern platforms use a combination of engagement-based and diversity-aware algorithms.

The goal of the study is to investigate the impact of the two recommendation strategies, namely engagement-based and diversity-aware, on user polarisation. The study aims to do this by examining social media interaction data and simulating the two recommendation strategies to measure their impact on user interactions and echo chambers.

II. Literature Review

Previous studies have explored how recommendation algorithms shape information consumption patterns. Facebook's algorithm lessens exposure to cross-ideological content, as Bakshy et al. showed [2]. Strong echo chamber effects were found by Cinelli et al. on a variety of social media platforms [3]. According to Ribeiro et al.'s analysis of YouTube's recommendation algorithm, users are drawn to more extreme content [4]. Misinformation exposure can exacerbate political polarization, as demonstrated by Guess et al. [5]. By purposefully adding different points of view to user feeds, diversity-aware recommendation systems may be able to reduce polarization, according to recent research [10][11]. Although previous research has shown the existence of echo chambers and algorithmic bias, the majority of these studies do not use computational models to quantitatively compare engagement-based and diversity-aware recommendation strategies. This research addresses this gap by integrating machine learning, NLP, and network analysis to provide a measurable evaluation of polarization.

III. Methodology

A. Data Collection

This research utilises an openly obtainable data set from sources like Kaggle, which includes user-generated content for social media sites (i.e., Reddit and Twitter). The data set contains user ID, post text, timestamps (the time when the post was made), and engagement metrics such as likes and replies. Data consists of 50k posts and 10k users, making for a representative sample of user and interaction behaviours on social media platforms. The data can be analysed to determine how users respond to various types/categories of content, as well as how users' opinions are created over time.

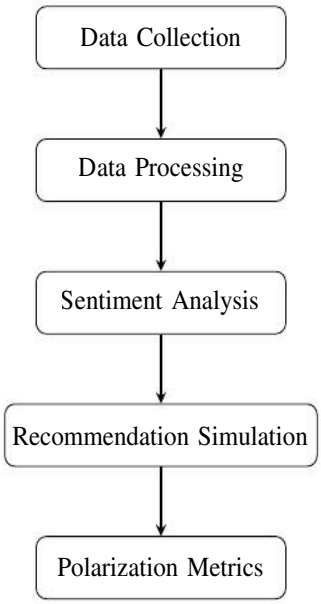

Fig. 1. Methodology flowchart of the proposed study

B. Data Processing

Data preprocessing occurred after the dataset had been collected. This process ensures that the dataset has been adequately cleaned and is prepared for subsequent analyses. The dataset must be free of noise and of high quality. First, the dataset's missing values were deleted. Next, all unneeded items were removed from the dataset, including but not limited to URLs, special characters, and any other symbols. All of the text was converted to lowercase; this ensured that all text was uniform. Basic natural language processing (NLP) techniques, such as tokenisation and stop word filtering, were also performed on the text data before the analysis. These techniques will simplify the text data and prepare it for any further analyses.

C. Sentiment Analysis

Sentiment analysis was utilised to assess people's sentiment based on the content of their posts. Each post received a sentiment score that indicated whether it was positive, negative, or neutral. In this way, we can determine how people's opinions are distributed across the board. By assessing sentiment as an indicator of public opinion toward a certain subject, it becomes easier to determine the extent to which opinions are different from one another and the degree to which they show bias towards one particular angle or another. We later used the sentiment scores to compute the following statistics regarding polarization: variance and average sentiment.

D. Recommendation Simulation

To study the effects of recommending systems, two methods of providing recommendations were simulated. The engagement-based recommendation system puts posts with more likes and interactions first. This method is an example of how real-world social networks rank content to produce maximum engagement from users. The diversity-based recommendation system selects posts randomly so that users will have access to a wider selection of content. This reduces bias among users when developing their opinions about posts online. The two methods of providing recommendations were used in order to see how recommending strategies influenced users' behaviours and opinions.

E. Polarization Metrics

The statistical measures of user polarization included an average sentiment score and variance. Sentiment variance indicates the distribution of opinions among users, indicating the degree of user polarization. A high sentiment variance indicates relatively high user polarization; a low sentiment variance indicates that users generally share a similar opinion. The average sentiment score tells us the direction of all user sentiment. Engagement-based recommendation systems and diversity-based recommendation systems were compared using both metrics. All experiments were implemented using Python, and results were visualised using basic data analysis libraries such as pandas and matplotlib.

IV. Results and Discussion

A. Comparison of Recommendation Approaches

This study was designed to examine the effect on user polarization of two types of recommendations - engagement-based and diversity-based. By using sentiment analysis on the data collected, multiple metrics related to user polarization were calculated, including the variance of sentiment in opinion and the average sentiment of opinion. In terms of the variance of sentiment in opinion, the engagement-based recommendation method had a much higher value than the diversity-based recommendation approach. This indicated that users' opinions were more polarised, meaning they accessed content that was congruent with their already held beliefs. Consequently, the opinions held by users were similar to one another, and an echo chamber was created. By contrast, the sentiment variance found in the diversity-based recommendation method was significantly lower. The lower variance means users had been exposed to more varied opinions, leading to a greater distribution of sentiment types. This resulted in less extreme polarization because of the introduction of differing viewpoints and opinions.

B. Quantitative Results

The engagement-based model has a larger variance than the diversity-based model based on the data presented in the table. This demonstrates that engagement-based recommendation systems create an increase in polarization amongst users. Whereas the lower variance of the diversity-based model indicates that it reduces polarization and creates a more even distribution of opinions than the engagement model.

TABLE I
Comparison of Polarization Metrics for Different Recommendation Models

Model	Sentiment Variance	Average Sentiment
Engagement-Based	0.65	0.20
Diversity-Based	0.38	0.10

C. Graph Analysis

This graph illustrates how user sentiment varies between two recommendation models, engagement-based and diversity-based, with the engagement-based model having a higher sentiment variance value of about 0.65 while the diversity-based model has a lower variance value of approximately 0.38. The fact that the engagement model has a higher sentiment variance value demonstrates that the opinions of users are much more diverse and polarized; therefore, they are much more likely to engage with content that supports their belief system, resulting in an echo chamber effect. The diversity-based model's lower variance indicates that the distribution of opinion is more balanced among users; therefore, exposing users to various kinds of content reduces polarization and encourages users to be more open-minded.

Furthermore, the average sentiment value of the two models is consistent with this trend, as the engagement model has slightly greater average sentiment than the diversity model, which supports the idea that engagement-based recommendation systems reinforce specific types of content. Overall, the correlation between these two recommendation models demonstrates how recommendation systems affect user behaviour by providing mixed recommendation strategies, thereby reducing echo chambers and polarization.

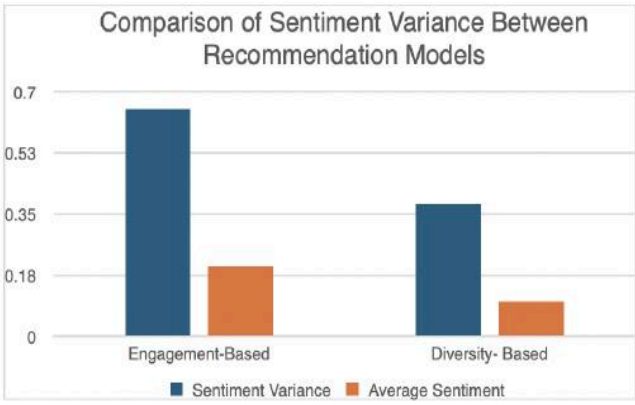

Fig. 2. Comparison of Sentiment Variance Between Recommendation Models

D. Discussion

The analysis of the findings reveals a significant impact of recommendation strategies on user behavior and the formation of opinions. Specifically, engagement-driven recommendation models are designed to optimize user interaction by proposing

content that corresponds with previously expressed user preferences. Although these strategies boost user activity, repeated exposure to similar materials reinforces established beliefs and can create echo chambers.

In contrast, the diversity-based recommendation model offers users a wider range of content. This approach gives users access to various sources, reducing bias and allowing for the exploration of different viewpoints. The results show that this model leads to a more balanced sentiment and lessens user polarization.

The study's results indicate a fundamental tension between user engagement and content diversity. Systems designed to maximize engagement significantly boost user interaction; however, they tend to recommend a narrow range of content, potentially fostering user polarization and the development of echo chambers. Conversely, systems that prioritize diversity expose users to a broader spectrum of content, which can encourage a wider array of viewpoints and mitigate user polarization, yet may simultaneously diminish engagement. Developing recommendation systems that find a balance between these two approaches is crucial. Combining engagement and diversity will improve the user experience and reduce the chances of echo chambers and polarization. These insights can help create socially responsible and effective recommendation system designs for social media platforms.

V. Conclusion

In this research, we explore how the recommendation system affects the growth of echo chambers and user polarization. Through an empirical study of both engagement-focused and diversity-oriented recommendation models, we find that how content is recommended to users significantly influences their responses and interactions with others. Recommendations based on engagement often suggest similar content, which plays a major role in increasing user polarization. As a result, users end up in echo chambers, only hearing a limited number of voices. On the other hand, in the diversity-aware recommendation algorithm, users get to hear a wider range of voices, which can make them want to find new and different opinions, leading to less user polarization. However, it mainly depends on the individual's preferences as to what kind of content they want to engage with. Even in diversity-aware recommendation platforms, users can still view one kind of content and ignore the other, thus resulting in echo chambers.

In general, this study's results suggest that both engagement and diversity should be considered when creating a comprehensive recommendation system, Currently platforms like Twitter and Reddit use both recommendation systems, which can lead to bringing in engagement and help minimize user polarization and reduce echo chambers. Engagement is important for the platform's growth, but the fact that users can personalize it a lot can have a greater effect on the diversity of information and how well users understand it. For future work, we can look into using better machine learning techniques and larger datasets to improve the evaluation and performance of these recommendation models.

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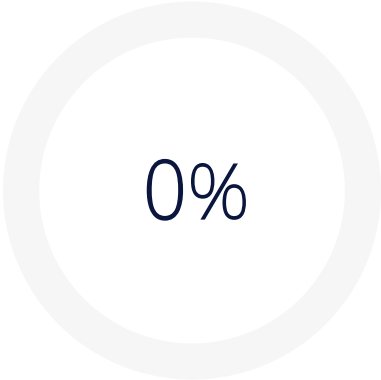

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